



ANALYSIS OF FLOW INSTABILITIES IN TURBOMACHINERY USING A NONLINEAR EIGENVALUE PROBLEM SOLVER DERIVED FROM HARMONIC BALANCE

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ABSTRACT

We present a computationally efficient method for determining the global stability of steady turbomachinery flows by solving a nonlinear, non-Hermitian eigenvalue problem (NEP). We demonstrate that the method computes several eigenpairs at a cost of only three to four steady flow calculations per eigenpair, making it practical for realistic three-dimensional configurations. The NEP is derived from a harmonic balance (HB) computational fluid dynamics (CFD) solver linearized about a steady base flow. The base nonlinear HB solver uses the Jameson–Schmidt–Turkel algorithm together with local time stepping, line-implicit residual smoothing, recursive multigrid, and Anderson mixing to efficiently compute nonlinear steady and periodic unsteady flows. Harmonic linearization of the HB residual equations defines an NEP of the form $A(\lambda)v = 0$, where the nonlinearity in $A(\lambda)$ can arise, for example, from two- or three-dimensional nonreflecting boundary conditions that are based on characteristic theory. The matrix–vector product $A(\lambda)v$ is computed efficiently using harmonic linearization, and a linearized form of the original fixed-point HB solver is used as a preconditioner in the eigensolver. Multiple eigenpairs of the non-Hermitian NEP are found using a soft deflation technique that does not require left eigenvectors. Computational results are presented for two test cases: vortex-shedding stability in low-Reynolds-number flow over a cascade of circular cylinders, and two- and three-dimensional unsteady flow in a compressor. These results illustrate the feasibility of applying the approach to realistic turbomachinery configurations in future global stability studies.

Keywords: Turbomachinery aerodynamics. Flow instabilities. Global stability analysis. Nonlinear eigenvalue problems.

Nomenclature

D Time-spectral time-derivative operator for periodic flow.
 m Number of retained frequencies: $m = 2N_h + 1$
 N_h Number of harmonics used in the harmonic-balance model
 N_t Number of time levels used in the harmonic-balance model

u Vector of conserved flow variables
 u^* Discrete solution vector sampled at multiple phase points (periodic state)
 U Steady base flow (equilibrium solution)
 v^* Small perturbation about the base flow
 $\hat{v}_{+1}$ First Fourier harmonic (complex amplitude) of the perturbation
 $\hat{v}_{-1}$ Complex conjugate Fourier harmonic of the perturbation
 λ Complex eigenvalue, $\lambda = \sigma + j\omega$
 σ Growth or decay rate of the mode
 ω Oscillation frequency (rad/s)
 j Imaginary unit, $j^2 = -1$
 R Nonlinear residual of the steady CFD system
 R^* Nonlinear harmonic-balance residual
 Q Spatial residual operator (discretized flux operator)
 Q^* Harmonic-balance spatial residual operator
 I Identity matrix
 $\otimes$ Kronecker product
 $\hat{R}$ First-harmonic projection of the residual
 Λ Diagonal matrix of Fourier wavenumbers
 E Discrete inverse Fourier transform matrix
 E^{-1} Discrete Fourier transform matrix
 ε Perturbation amplitude for finite-difference approximations
 τ Pseudo-time variable for iterative convergence
 ρ_T Inlet stagnation density
 p_T Inlet stagnation pressure
 p Static pressure
 μ Dynamic viscosity
 γ Ratio of specific heats
 R Specific gas constant
 D Cylinder diameter
 G Cylinder-to-cylinder spacing
 Re_D Reynolds number based on cylinder diameter
 x, y, z Cartesian coordinates
 x, r, θ Cylindrical coordinates
 k Circumferential wavenumber

1 Introduction

In this paper, we describe a straightforward and efficient method for performing global stability analysis of three-dimensional turbomachinery flows. The approach is based on a harmonic balance (HB) formulation and leads to a nonlinear, non-Hermitian eigenvalue problem. Rather than forming and storing Jacobian large matrices, the method uses matrix-free techniques to define the required matrix–vector products directly from the underlying CFD solver. This makes it possible to compute a small number of dominant eigenpairs efficiently, without major changes to the existing solver. The goal is to provide a practical tool for analyzing flow instabilities—such as rotating stall and non-synchronous vibration—in realistic turbomachinery configurations.

Rotating flow instabilities, such as non-synchronous vibration (NSV) and rotating stall in turbomachinery, limit the performance of fans and compressors and, in severe cases, can lead to catastrophic engine failure. In this paper, we focus on global stability analysis—the calculation of eigenvalues and eigenmodes that describe the onset of these global rotating flow instabilities. While predicting the fully developed nonlinear response of an instability is essential for assessing its impact on performance or aeromechanical behavior, eigenvalues and eigenmodes offer fundamental insight into the underlying physics. Global stability analysis can identify loss-of-stability trends before the boundary is crossed and provides a measure of stability margin.

Rotating flow instabilities in turbomachinery—including rotating stall and non-synchronous vibration (NSV) [1–6]—remain a persistent challenge in the design and safe operation of aircraft turbofan engines. Rotating stall involves stalled regions (“stall cells”) that rotate around the rotor at roughly 50–70% of rotor speed and may take the form of long-wavelength disturbances spanning multiple blade passages or highly localized “spikes” that propagate from blade to blade [2, 7]. Stall can progress to surge, a violent axisymmetric oscillation of the flow through the compressor. For an excellent historical review of experimental, theoretical, and computational stall research, see Day [1].

Non-synchronous vibration (NSV) is also driven by a rotating flow instability, but it is typically milder than rotating stall. If the instability frequency lies close to a blade’s natural frequency, it can *lock on* to the structural motion [8, 9], reinforcing vibration and leading to high-cycle fatigue. Blade motion may also organize otherwise low-amplitude, chaotic separated-flow vibration (SFV) into coherent, high-amplitude NSV.

Most stability analyses of turbomachinery instabilities have used either low-order, physics-based models built from compressor pumping characteristics (e.g., [10, 11]) or expensive direct computational simulations using RANS models (e.g., Ref. [2, 3]). Gourdain et al. [3] performed full annulus simulations of the onset and development of rotating stall in a compressor stage (rotor/stator) experimental facility using a RANS analysis. The simulation required roughly 4,000 CPU hours and successfully predicted the mass flow at stall onset, though it did not correctly predict the number of stall cells or their rotation speed. Pullan et al. [2] studied the onset and development of spike instabilities in the NASA E³ Rotor B and the Cambridge low-speed compressor using full-rotor RANS simulations.

Tharreau et al. [12] modeled NSV in the ultra-high by-

pass ratio (UHBR) ECL5/CATANA fan using a time-accurate URANS code with about 200 time steps per blade-passing period and a bi-directionally coupled CFD/CSD model using the first three finite-element structural blade modes. They computed how vortical disturbances form, propagate, and lock in with blade modes, finding a rotating instability traveling at about 59% of rotor speed (in the absolute frame).

An alternative to computationally expensive time-domain approaches are frequency-domain methods, including time-linearized [13–15] and harmonic balance [16] techniques. These computational methods have revolutionized unsteady flutter and forced-response analyses for turbomachinery: dramatically reducing cost, improving accuracy, and enhancing understanding. However, frequency-domain techniques have not been as fully developed for analyzing rotating flow instabilities, including NSV, rotating stall, and problems involving larger-scale flow separations.

At the heart of rotating flow instabilities are unstable, small-disturbance *linear* instabilities that grow into fully developed *nonlinear* limit cycles. Several studies have computed linear global stability eigenvalues for the onset of rotating instabilities in turbomachinery. However, to the authors’ knowledge, all this work has used *bi-global* rather than *tri-global* analysis. That is, the eigen-problem is reduced from three to two dimensions to manage computational cost—either by ignoring a third dimension or using model reduction.

For example, Parker’s work on acoustic resonances and their effect on flow-induced vibrations of turbomachinery blading was among the first to examine fluid-dynamic eigenmodes. He computed a small set of acoustic modes for low-Mach-number flow over a cascade of unstaggered, two-dimensional flat plates by numerically solving the acoustic wave equation [17] in the axial-circumferential (x, θ) plane. He proposed that these modes could couple with Strouhal-number shedding near frequency crossings, producing large aerodynamic excitation forces—although he did not directly solve the coupled acoustics/shedding problem, instead examining frequency coincidences.

In later work, Parker solved the axisymmetric acoustic wave equation for low-Mach-number flow in a compressor annulus with radii varying from inlet to exit and for prescribed circumferential wave numbers (nodal diameters) [18]. He built a physical model of a compressor annulus with hard-wall terminations at the inlet and exit and experimentally measured resonant frequencies, finding good agreement with predictions for lower-nodal-diameter modes. Parker suggested that these modes could lock in to rotor–stator interaction excitations near rotor–stator and acoustic-mode frequency crossings.

More recently, several investigators have performed bi-global stability analyses of rotating stall and other flow instabilities. Xu et al. [19] performed an eigenanalysis of a two-dimensional annular slice of Rotor 67 to predict stall inception. Glazkov et al. [20, 21] carried out eigen-analyses of an idealized two-dimensional compressor to investigate subsonic flow instabilities on very fine two-dimensional meshes, again ignoring the radial dimension to reduce problem size.

Others have used reduced-order modeling. Several studies have modeled the onset of rotating stall in axial compressor rotors [4, 22] and centrifugal compressors [5] using 2D eigen-

analysis in the axial/radial plane (x, r) , linearizing about the circumferentially averaged steady flow. Flow perturbations were assumed to vary like $\exp(jk\theta)$ in the (third) circumferential dimension, where k is a prescribed wavenumber. Blade forces were replaced with circumferentially distributed body forces. Hu et al. [6] found eigenvalues and eigenvectors of flow instabilities in the vaneless diffusers of centrifugal compressors, again assuming Fourier behavior in θ , though their work was restricted to incompressible flow and did not fully linearize turbulence models. They obtained both left and right eigenvectors, enabling identification of so-called ‘wavemaker’ regions.

In this paper, we present a computational method for fully three-dimensional global stability analysis of unsteady flows in turbomachinery by solving a nonlinear eigenvalue problem (NEP) [23] for small disturbance flow in a blade row. The method uses a conventional harmonic balance flow solver to define the NEP and employs the CFD solver as a preconditioner to numerically compute stability eigenvalues and corresponding right eigenvectors. The analysis is implemented using a matrix-free eigen-solver, with the eigenvalue problems formulated as an extension to an existing harmonic balance analysis.

2 Technical Approach

2.1 Nonlinear Harmonic Balance Solver

We begin by spatially discretizing the Reynolds-averaged Navier–Stokes equations, including the Spalart–Allmaras [24, 25] one-equations turbulence model for turbulent flows. For the present study, we discretize the governing equations on a block-structured mesh using the cell-centered finite-volume Jameson–Schmidt–Turkel (JST) scheme [26, 27] This yields a large system of nonlinear ordinary differential equations of the form

$$\frac{du}{dt} + Q(u) = 0, \quad (1)$$

where u is the vector of conserved flow variables, and $Q(u)$ contains the spatially discretized terms.

To compute unsteady, periodic solutions—such as those found in turbomachinery flutter, rotating stall, or non-synchronous vibration (NSV)—we apply the harmonic balance (HB) method [16]. In this approach, we assume the solution is periodic in time with a known frequency ω , and solve for the state at N_t discrete time points within one period of oscillation. The resulting solution residual is

$$R^*(u^*, \omega) = (D \otimes I)u^* + Q^*(u^*, \omega) = 0, \quad (2)$$

where u^* is the stacked periodic solution at the N_t discrete time instances, D is a spectral time-derivative matrix, and $\otimes$ denotes the Kronecker product with the identity matrix I . (The superscript “*” here refers to the stacked time-spectral state, not complex conjugation.)

In the original nonlinear harmonic balance method, the solution is assumed to lie on a limit cycle and is approximated by a truncated Fourier series. For example, if we retain a single harmonic ($N_h = 1$), then the number of time instances is

$N_t = 2N_h + 1 = 3$, and the unsteady solution at time index m is given by the inverse Fourier transform:

$$u_m^* = \sum_{n=-1}^1 \hat{u}_n \exp\left(\frac{2\pi jmn}{N_t}\right), \quad (3)$$

where $\hat{u}_n$ are the complex Fourier coefficients and $m = 0, 1, 2$ indexes the time slices.

The corresponding time derivative at each time instance is

$$\frac{du_m^*}{dt} = j\omega \sum_{n=-1}^1 n \hat{u}_n \exp\left(\frac{2\pi jmn}{N_t}\right). \quad (4)$$

This relationship can be written in matrix form as

$$\frac{du^*}{dt} = (D \otimes I)u^*, \quad (5)$$

where u^* is the vector of solution snapshots at each time slice and D is the discrete time-spectral derivative matrix, constructed as

$$D = E \Lambda E^{-1}. \quad (6)$$

Here E is the discrete Fourier matrix $E_{mn} = \exp(2\pi jmn/N_t)$, and E^{-1} is the corresponding discrete Fourier transform (DFT) matrix given by $E^{-1} = \frac{1}{N_t} E^H$. The matrix Λ is diagonal matrix with Fourier wavenumbers ω , that is, $\Lambda = \text{diag}(-j\omega, 0, j\omega)$.

The formulation given by Eqs. (2) through (6) replaces time-accurate time-domain integration with a system of nonlinear equations R^* representing the periodic solution directly. Note that $R^*(u^*, \omega)$ is nonlinear in the flow solution u^* and potentially in the frequency ω as well. While the spectral operator D is linear in ω , nonreflecting boundary conditions may produce terms nonlinear in λ (or ω for the original nonlinear HB) in the flux terms $Q^*(u^*, \omega)$. See, for example, Ref. [28, 29].

We solve Eq. (2) using pseudo-time marching, introducing a fictitious time variable τ to drive the solution to steady state:

$$\frac{du^*}{d\tau} + R^*(u^*, \omega) = 0, \quad (7)$$

so that the desired periodic solution is recovered when the pseudo-time term vanishes (i.e., when Eq. (2) holds). A four-stage Runge–Kutta-like method is used for pseudo-time marching, accelerated with local time stepping, residual averaging, and recursive multigrid [26, 27]. We denote the recursive multigrid cycle as V^k , where k specifies the number of recursive traversals at each level. Thus, V^1 is a standard V-cycle, V^2 is a W-cycle, and higher values (e.g., V^3) correspond to deeper W-type cycles. In practice, we typically use V^2 or V^3 . This process defines a fixed-point iteration of the form,

$$u^* := N^*(u^*, \omega), \quad (8)$$

which is repeated until convergence.

For flow instabilities, the limit cycle frequency ω is not generally known *a priori*. In these situations, we update ω at each iteration after updating u^* . We assume that the estimate for u^* is fixed and perform a Newton–Gauss correction that minimizes the norm of the residual with respect to ω . The update for the

frequency is:

$$\omega := \omega - \frac{R_\omega^* R^*}{R_\omega^* R_\omega^*}, \quad (9)$$

where R_ω^* is the derivative of the residual with respect to frequency, which can be computed using a finite-difference operator (details in the next section).

Because the system is time-shift invariant, the phase of the solution is arbitrary, just as the phase of the eigenvector of a standard eigenvalue problem is arbitrary. However, due to nonlinear effects, the *magnitude* of the solution is not arbitrary and is uniquely determined by the nonlinear balance of the governing equations. In practice, the frequency update and residual minimization implicitly select a consistent phase, though an explicit phase condition could also be imposed. Equations (8) and (9) are iterated until convergence.

This harmonic-balance framework allows us to compute large-amplitude, nonlinear, time-periodic solutions—such as limit cycle oscillations (LCOs)—in computational times that scale like the time required to compute a single steady flow field times the number of time levels used in the harmonic-balance analysis. The number of retained frequencies is typically $m = 2N_h + 1$, where N_h is the number of harmonics in the model, and $N_t = m$ in the collocation scheme used here. The number of harmonics needed to model nonlinearities in turbomachinery is typically three or less if one is interested in predicting the mean performance or the unsteady response at the fundamental frequency.

The method outlined above can often compute nonlinear, unsteady, periodic flows, including limit cycle oscillations (LCOs). In some cases the flow may be chaotic or aperiodic, and no stable periodic solution exists.

Having described the standard harmonic balance approach for solving large-amplitude periodic flows, we now turn our attention to the formulation and solution of the small disturbance global stability analysis of unsteady flows using harmonic linearization of a nonlinear harmonic balance code.

2.2 Harmonic Linearization

In this section, we formulate a matrix-free nonlinear eigenvalue problem (NEP) to compute the small-disturbance eigenvalues and corresponding eigenmodes that characterize the onset of flow instabilities about a steady base flow. This NEP is derived directly from the harmonic-balance formulation described above. The resulting eigenpairs provide physical insight into the dominant instability mechanisms and quantify the stability margin, even for linearly stable modes.

To formulate the eigenvalue problem, we start with the nonlinear harmonic balance (HB) residual equation,

$$R^*(u^*, \lambda) = (D \otimes I) u^* + Q^*(u^*, \lambda), \quad (10)$$

where u^* is the discrete solution vector sampled at several phase points within a single oscillation period, and D is the spectral time-derivative operator. Note that the D operator now depends on $\lambda = \sigma + j\omega$, not just $j\omega$.

We assume the unsteady solution consists of a time-invariant steady flow U , plus a small perturbation with exponential behav-

ior. That is,

$$u^* = U + \frac{1}{2} \text{Re} \left[v e^{\lambda t^*} \right], \quad (11)$$

where $\lambda = \sigma + j\omega$ is a complex eigenvalue with oscillation frequency ω and growth rate σ

Substituting into the harmonic balance equations yields

$$R^*(u^*, \lambda) = (D \otimes I) u^* + Q^*(u^*, \lambda), \quad (12)$$

where the periodic spectral operator D is now frequency-dependent, and given by

$$D = E \Lambda E^{-1}, \quad (13)$$

with $\Lambda = \text{diag}(\sigma - j\omega, 0, \sigma + j\omega)$, and E and E^{-1} denoting the discrete Fourier transform and its inverse, respectively.

Next, we apply a discrete Fourier transform to the residual R^* , yielding

$$\hat{R} = E R^*. \quad (14)$$

The mean residual $\hat{R}_0$ is zero by construction (for small disturbances), and $\hat{R}_{+1}$ and $\hat{R}_{-1}$ are complex conjugates. We therefore seek

$$\hat{R}_{+1}(v, \lambda; U) = 0. \quad (15)$$

Linearizing this expression about the for small disturbances v_{+1} yields the eigenvalue problem,

$$\hat{R}_v(\lambda; U) v = 0, \quad (16)$$

where $\hat{R}_v$ is the Jacobian of the first-harmonic residual with respect to v , evaluated at the steady base flow U and parameterized by the complex frequency λ . (We have dropped the subscript $+1$ on v and $\hat{R}$ for clarity.)

Equation (16) is our desired nonlinear eigenvalue problem. Nonlinear eigenvalue problems are linear in the mode shape, but nonlinear in λ [23]. For simple (reflective) boundary conditions, λ only appears linearly in the time-spectral term D . However, implementation of non-reflecting boundary conditions λ can result in nonlinear terms appearing in the flux terms.

As a practical matter, in our computations we never explicitly form the Jacobian matrix $\hat{R}_v$. Instead, we use a matrix-free finite-difference approach to evaluate the matrix-vector product $\hat{R}_v(\lambda; U) v$ for a given eigenvector estimate v , i.e.,

$$\hat{R}_v(U, \lambda) v \approx \frac{1}{\varepsilon} [\hat{R}(U + \varepsilon v, \lambda) - \hat{R}(U, \lambda)], \quad (17)$$

where ε is a small real parameter. Because the base flow satisfies $\hat{R}(U, \lambda) = 0$, this expression simplifies to

$$\hat{R}_v(U, \lambda) v \approx \frac{1}{\varepsilon} \hat{R}(U + \varepsilon v, \lambda). \quad (18)$$

Since the dependence on v is linear (and hence analytic), the finite-difference approximation in Eqs. (17)–(18) is well defined. This matrix-free formulation avoids forming the full Jacobian and requires only a single evaluation of the nonlinear residual per perturbation direction.

We note that the Jacobian $\hat{R}_v$ is non-Hermitian, and therefore its left and right eigenvectors are not equal. One disadvan-

tage of the matrix-free approach is that we do not have direct access to the Jacobian or its adjoint, and therefore cannot compute the left eigenvectors. Left eigenvectors are often used to deflate previously computed eigenpairs when seeking multiple eigenpairs; without the left vectors, extracting successive eigenvalues becomes more difficult.

2.3 A Simple Eigensolver

If a single eigenpair is desired, the simplest approach is to simply linearize the nonlinear harmonic balance solver using harmonic linearization. The steady equilibrium flow is found by solving The steady equilibrium flow is found by solving

$$R(U) = 0, \quad \text{with} \quad R, U \in \mathbb{R}^n, \quad (19)$$

$$U := N(U), \quad (20)$$

Similarly, the nonlinear harmonic balance solution is given by

$$R^*(u^*, \omega) = 0, \quad \text{with} \quad R^*, u^* \in \mathbb{R}^{n \times m}, \quad (21)$$

$$u^* := N^*(u^*, \omega), \quad (22)$$

where m is the number of retained time levels (or modes), and N^* is the corresponding fixed-point operator in the harmonic balance framework.

Finally, the nonlinear eigenvalue solution is defined by

$$\hat{R}_v(v, \lambda; U)v = 0, \quad \text{with} \quad \hat{R}_v, v \in \mathbb{C}^n, \quad (23)$$

$$v := \hat{N}_v(\lambda; U)v \quad (24)$$

The product $\hat{N}_v(\lambda; U)v$ is computed using a similar matrix-free harmonic linearization approach as was applied to the residual $\hat{R}$ in the previous section.

To find a single eigenpair, we iteratively solve Eq. (24) to estimate the eigenvector. After each iteration, the eigenvector is normalized, and the corresponding eigenvalue $\lambda = \sigma + j\omega$ is updated using a Newton–Gauss step. Specifically, we minimize the squared norm of the first harmonic of the residual with respect to the eigenvalue components σ and ω by solving

$$\min_{\sigma, \omega} \|\hat{R}(\hat{v}_{+1}, \lambda)\|^2. \quad (25)$$

In principle, one could treat λ as a complex variable and apply a Newton update using a Taylor expansion:

$$\hat{R}(\lambda + \Delta\lambda) \approx \hat{R}(\lambda) + \hat{R}_\lambda \Delta\lambda, \quad (26)$$

which yields the formal complex update:

$$\Delta\lambda = -\frac{\hat{R}_\lambda^H \hat{R}}{\hat{R}_\lambda^H \hat{R}_\lambda}. \quad (27)$$

This approach, however, assumes that $\hat{R}$ is analytic in λ , which may not be true in practice—for example, due to non-analytic boundary conditions at the far field.

To refine the eigenvalue in a more robust way, we instead treat λ as a real two-dimensional variable and minimize the

residual norm using the real-valued Newton–Gauss update,

$$\text{Re} \begin{bmatrix} \hat{R}_\sigma^H \hat{R}_\sigma & \hat{R}_\sigma^H \hat{R}_\omega \\ \hat{R}_\omega^H \hat{R}_\sigma & \hat{R}_\omega^H \hat{R}_\omega \end{bmatrix} \begin{bmatrix} \Delta\sigma \\ \Delta\omega \end{bmatrix} = -\text{Re} \begin{bmatrix} \hat{R}_\sigma^H \hat{R} \\ \hat{R}_\omega^H \hat{R} \end{bmatrix}. \quad (28)$$

In practice, the derivatives $\hat{R}_\sigma$ and $\hat{R}_\omega$ are approximated using finite differences. Because the residual $\hat{R}$ is linear in λ (except for nonlinear dependencies introduced through nonreflecting boundary conditions), the finite-difference step size need not be excessively small. For robustness, Eq. (28) is typically under-relaxed with a factor in the range 0.1–0.5.

This simple algorithm for finding a single eigenpair is shown in Algorithm 1. While not guaranteed, the least stable eigenpair is often the one that naturally emerges, especially when a reasonable initial guess for the eigenvalue is provided.

Algorithm 1

Matrix-Free Nonlinear Eigenvalue Solver (Right-Eigenvector Only)

- 1: **Input:** Initial guess λ , tolerance τ
 - 2: **Initialize:** Random vector $v \leftarrow \text{rand}()$, normalize $v \leftarrow v/\|v\|$
 - 3: **while** not converged **do**
 - 4: Compute residual-based directional update:

$$v \leftarrow \hat{N}(\lambda; U)v$$
 - 5: Normalize: $v \leftarrow v/\|v\|$
 - 6: Update eigenvalue λ using a Newton–Gauss step
 - 7: **end while**
 - 8: **Output:** Approximate eigenpair (λ, v)
-

Algorithm 2

Matrix-Free Nonlinear Eigenvalue Solver with Soft Deflation

- 1: **Initialize:** $V \leftarrow []$ ▷ Empty deflation basis
 - 2: **for** each mode $j = 1, 2, \dots$ **do**
 - 3: **Initialize:**
 - 4: $\lambda \leftarrow$ initial guess
 - 5: $v \leftarrow$ random vector
 - 6: Orthogonalize: $v \leftarrow (I - VV^H)v$
 - 7: Normalize: $v \leftarrow v/\|v\|$
 - 8: **while** not converged **do**
 - 9: Compute residual-based correction:

$$\Delta v \leftarrow \hat{N}(\lambda; U)v - v$$
 - 10: Apply soft deflation and orthogonal update:

$$\Delta v \leftarrow (I - v v^H)(I - \beta V V^H)\Delta v$$
 - 11: Update and normalize:

$$v \leftarrow v + \Delta v, \quad v \leftarrow v/\|v\|$$
 - 12: Refine eigenvalue λ using a Newton–Gauss step
 - 13: **end while**
 - 14: **Lock converged mode:**
 - 15: $v_j \leftarrow v, \quad \lambda_j \leftarrow \lambda$
 - 16: Expand deflation basis: $V \leftarrow [V \quad v_j]$
 - 17: **end for**
 - 18: **Output:** Eigenpairs $\{(\lambda_j, v_j)\}$
-

2.4 Solution Technique for Multiple Eigenpairs

To compute multiple eigenpairs, deflation is typically used to suppress modes that have already converged. In our case, the eigenvalue problem is not only nonlinear but also non-Hermitian. Because we use a matrix-free approach, the Jacobian is never explicitly formed, and the Jacobian adjoint is not available. Thus, the corresponding left eigenvectors—required for standard deflation techniques—cannot be computed and are unavailable for use in the deflation process.

Fortunately, in global stability analysis, we are usually interested in only a small number of the least stable modes, which are often well-separated in the complex plane. For such problems, so-called *soft deflation* is often sufficient to suppress previously converged eigenmodes and enable convergence to new ones.

Using soft deflation, Eq. (24) is modified to suppress—though not completely eliminate—the component of the eigenvector correction in the direction of previously converged eigenvectors. Specifically,

$$v := v + (I - v v^H)(I - \beta V V^H)(\hat{N}_v(\lambda; U)v - v) \quad (29)$$

where V is the matrix whose columns are the previously converged eigenvectors, and $\beta \in (0, 1)$, typically chosen near 0.95. The soft deflation term $(I - \beta V V^H)$ reduces the influence of prior modes, thereby slowing their numerical growth relative to the next unconverged mode and steering the iteration toward a new eigenpair. To maintain numerical stability and prevent growth in the component of the update Δv along the current iterate v , we include the projection $(I - v v^H)$. While this term is theoretically redundant due to the normalization of v after the update, in practice it helps mitigate the accumulation of numerical error, especially in early iterations when updates can be relatively large.

The advantages of the global stability analysis described in this section are several. (1) The eigenvalue problem $\hat{R}_v(\lambda)v = 0$ can be easily defined from an existing harmonic balance code using harmonic linearization as a wrapper, with reasonable computational efficiency. (2) The CFD/HB iterator developed to efficiently solve complex three-dimensional unsteady flows serves as an effective preconditioner for computing eigenpairs. Typically, the time required to compute a single eigenpair is only about three to four times the cost of computing the underlying steady flow solution. (3) The method accommodates nonlinear eigenvalue problems directly. A number of papers in the literature impose reflecting boundary conditions at the far field to force the global stability analysis into the form of a standard or generalized eigenvalue problem, so as to be compatible with available software packages. Our approach avoids such simplifications and preserves physically consistent boundary modeling. (4) The matrix-free formulation avoids explicit construction or storage of the Jacobian, enabling application to large-scale CFD problems with modest memory overhead.

3 Computational Results

We now demonstrate the proposed method using three representative problems. First, we consider classical two-dimensional, low-Reynolds-number vortex shedding from an isolated circular

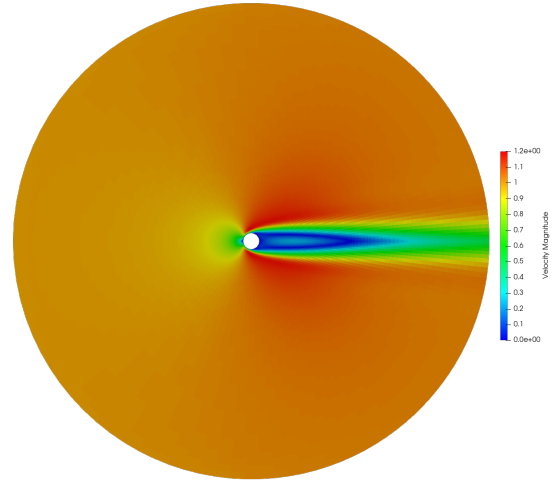


FIGURE 1: Computed unstable steady equilibrium flow over a cylinder in crossflow at a Reynolds number $Re_D = 100$. Contours of velocity magnitude are shown.

cylinder, to verify that the approach reproduces well-known instability mechanisms and to assess its efficiency. Second, we examine vortex shedding from a two-dimensional array of cylinders, which shares some features with turbomachinery cascades and provides a useful intermediate test case. Finally, we apply the method to Purdue Rotor 2, a realistic three-dimensional turbomachinery configuration, to illustrate its application to practical problems.

3.1 Vortex Shedding

3.1.1 Vortex Shedding about an Isolated Cylinder

To demonstrate the method, we first consider the classic problem of two-dimensional vortex shedding about a cylinder in crossflow. We compute the steady equilibrium flow field using our HB flow solver with a single time level. As an example, consider the case with Reynolds number $Re = 100$. A structured computational grid with an outer radius of 15 diameters is used. The grid consists of $128 \times 128 \times 2$ computational cells, for a total of 32,768 cells. For this example, we use five levels of multigrid with a simple V-cycle.

Figure 1 shows the computed steady equilibrium flow. Note that for this Reynolds number the flow is physically unstable. However, with local pseudo-time-stepping and multigrid acceleration, the steady flow solver is *computationally* stable, providing the equilibrium state about which we perform the eigenanalysis.

Having computed the steady flow, we next compute the least stable eigenpair using the method described in Section 2.3. Figure 2 shows the convergence history of the least stable eigenvalue and the norm of the solution residual. An initial guess of the eigenvalue is provided, and the perturbation flow field is seeded with random numbers. The eigensolver is first run for 20 warmup iterations to allow an approximate eigenvector to emerge before updating the eigenvalue. The eigenvalue converges rapidly, reaching engineering accuracy in about 100 iterations, requiring approximately 467 seconds on a Mac Studio M3 using a single processor. For comparison, the steady flow calculation required

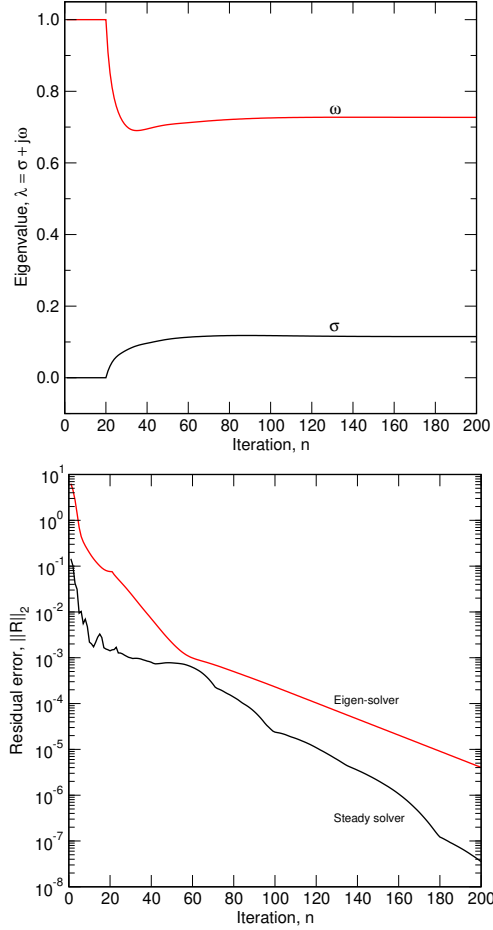


FIGURE 2: Left: convergence history of the real and imaginary parts of the least unstable eigenvalue for the onset of vortex shedding at $Re_D = 100$. Right: convergence of residual norms.

about 100 iterations and 125 seconds to achieve a four-order-of-magnitude reduction in the residual norm. The cost to compute a single eigenpair is therefore only 3.7 times that of a steady computation.

Shown in Fig. 3 is the corresponding eigenmode for the eigenvalue just computed. Plotted are the real and imaginary parts of the x and y components of velocity.

3.1.2 Vortex Shedding from an Array of Cylinders

The computed non-dimensional eigenvalues are plotted in Fig. 4 (dimensional eigenvalues multiplied by the diameter and divided by the freestream velocity) for a range of Reynolds numbers. For comparison, eigenvalues computed by Crouch et al. [30] are also shown. Note the very good agreement between the eigenvalues computed using the present method and those of Crouch et al.

Also shown in the figure is the *nonlinear* harmonic balance (HB) solution computed using one harmonic on a $256 \times 256 \times 2$ cell grid with the same HB code used within the eigensolver. The measured limit-cycle frequency for vortex shedding about a cylinder in a water tow tank is also included [31]. There is generally good agreement with the experimental data. Note that no limit-cycle HB solution exists for Reynolds numbers below

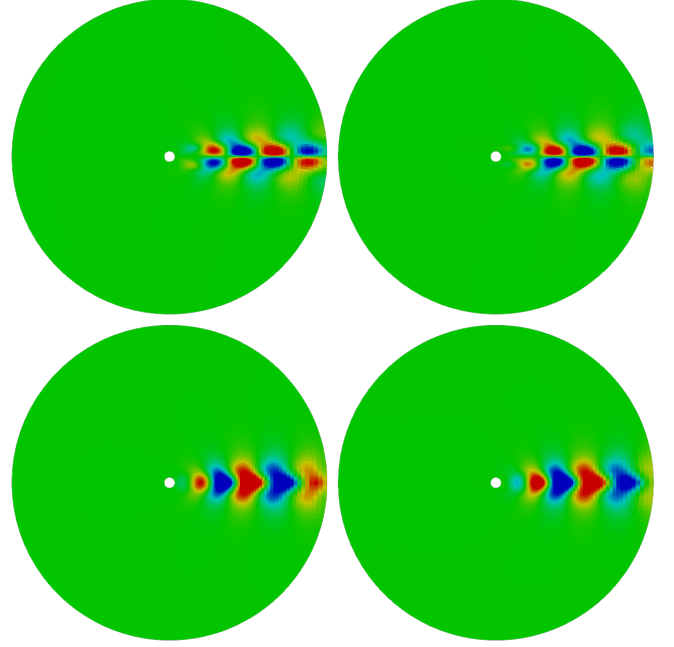


FIGURE 3: Computed eigenmodes of vortex shedding at $Re_D = 100$. Top: real and imaginary parts of the x -component of perturbation velocity. Bottom: y -component.

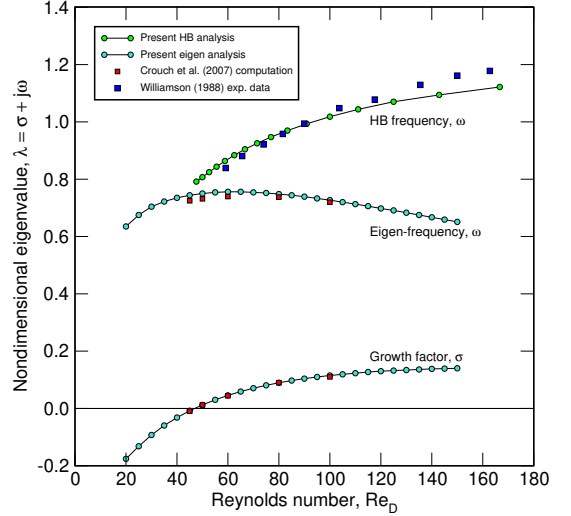


FIGURE 4: Non-dimensional shedding frequency of a cylinder in low-Reynolds-number crossflow, computed using the present linear stability analysis and a nonlinear HB solver with one harmonic, and compared with previous experimental and computational results.

the onset of linear instability, $Re_D \approx 47$. Furthermore, the computed HB frequency is always larger than the small-disturbance frequency, indicating that the physical system exhibits characteristics of a stiffening spring.

These computational results, although for a relatively simple example, illustrate several important features of HB computations and the present nonlinear eigenvalue analysis: (1) The eigen-analysis computes both frequency and growth factor, and

the fully developed LCO frequency can differ significantly from the onset frequency. This behavior is also observed in rotating instabilities in turbomachinery. (2) The number of iterations required to compute eigenvalues to engineering accuracy is about the same as for the steady flow solver. (3) Local time stepping, multigrid acceleration, and other techniques commonly used to accelerate steady flow convergence can stabilize computations of physically unstable flows. Such solvers should therefore not be used as a surrogate for a proper stability analysis.

We next consider a simple model problem to demonstrate the method's ability to compute eigenmodes and eigenvalues of flow instabilities. Specifically, we look at low-Reynolds-number flow over a cascade of circular cylinders. The nondimensional flow parameters are: inlet stagnation density and pressure $\rho_T = 1$ and p_T , exit static pressure $p = 0.95$, viscosity $\mu = 0.005$, and gas properties $\gamma = 1.4$ and $R = 1.0$. The cylinder diameter is $D = 2.0$, and the cylinder-to-cylinder spacing is $G = 5.0$. For these conditions, the flow is nearly incompressible with an inlet velocity of 0.1910 resulting steady flow Reynolds number Re_D of about 76. For this case, there are a total of 30,720 computational cells. The problem is two-dimensional, but our three-dimensional code requires at least two cells in each direction, so each of the two-dimensional grid layers has 15,360 cells (see Figure 5).

Figure 5 shows the computed steady, unstable equilibrium flow field and the computational grid. In each passage, contours of velocity magnitude, entropy, and pressure are shown from top to bottom. Note the long wake of velocity deficit and entropy rise behind the cylinder.

Physically, the flow is unstable at this Reynolds number, but the eigenvalues of the *physical system* differ from those of the *computational steady flow solver*. The solver's eigenvalues are all stable, meaning the magnitude of the computational growth factors of the computational eigenmodes satisfy $|Z| \leq 1$. Sta-

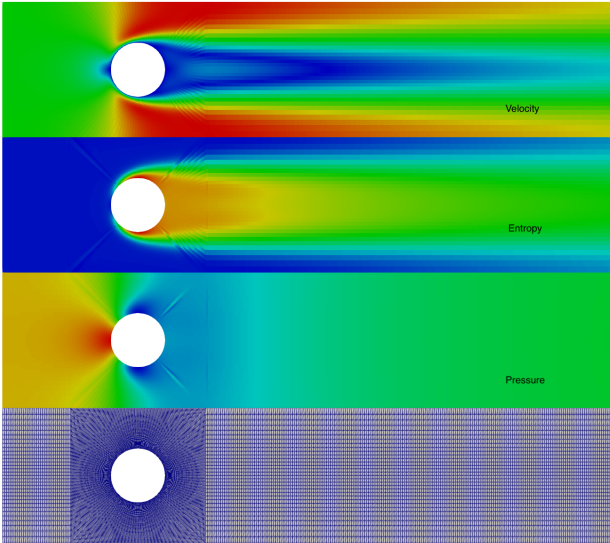


FIGURE 5: Computed (unstable) steady equilibrium flow for low-Reynolds-number flow over a cylinder cascade. Shown in each passage from top to bottom are contours of velocity magnitude, entropy, and pressure, and the computational grid.

bility is achieved through the combined stabilizing effects of Runge–Kutta integration with local time stepping, line-implicit residual averaging, recursive multigrid acceleration, and sometimes Anderson mixing [32]. (GMRES, proper orthogonalization techniques [33], and the recursive projection method [34] for stabilizing fixed-point iterations could also be used.)

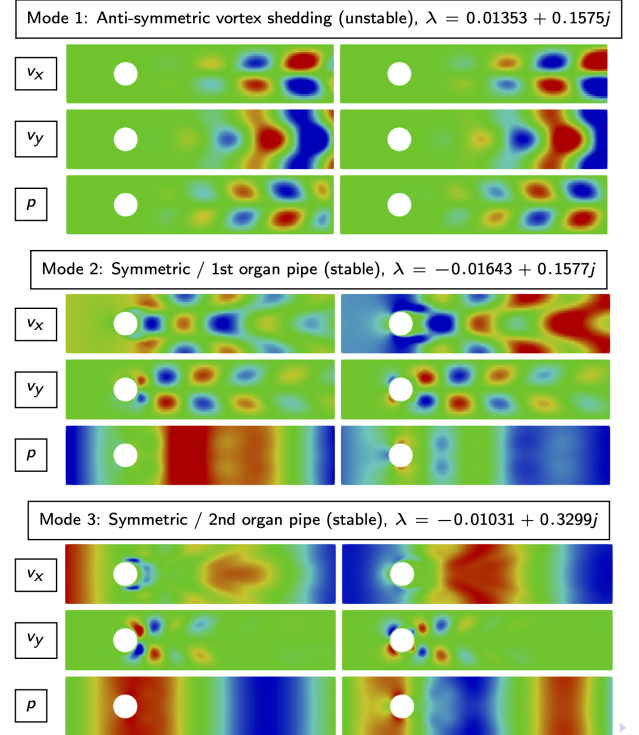


FIGURE 6: First three non-spurious eigenmodes for the cylinder cascade case. Mode 1 is the unstable vortex-shedding mode; modes 2 and 3 are stable organ-pipe modes.

Having computed the steady flow, we now compute a few of the low-frequency eigenpairs using the method described in Algorithm 2. Figure 7 shows the first five computed eigenvalues, and also the convergence history for this computation.

Note two points: (1) The first four eigenpairs show linear convergence over several orders of magnitude, while the fifth stalls at a relatively large residual. (2) Eigenvalue four appears to be a duplicate of eigenvalue one. Such stalled convergence of later eigenpairs and the appearance of spurious duplicates are common when using soft deflation for non-Hermitian problems, as the lack of left eigenvectors prevents full removal of previously converged modes.

Figure 6 shows the first three non-spurious eigenmodes. The first mode is the fundamental vortex-shedding mode and is our primary mode of interest. The real part of its eigenvalue λ is positive, indicating instability, while all other computed eigenmodes are stable. The frequency for this mode, $\omega = 0.1353$, corresponds to the onset of the instability.

Figure 9 also shows the second and third modes, which are best described as organ-pipe modes. In this example, we use reflecting boundary conditions—total pressure and density fixed at

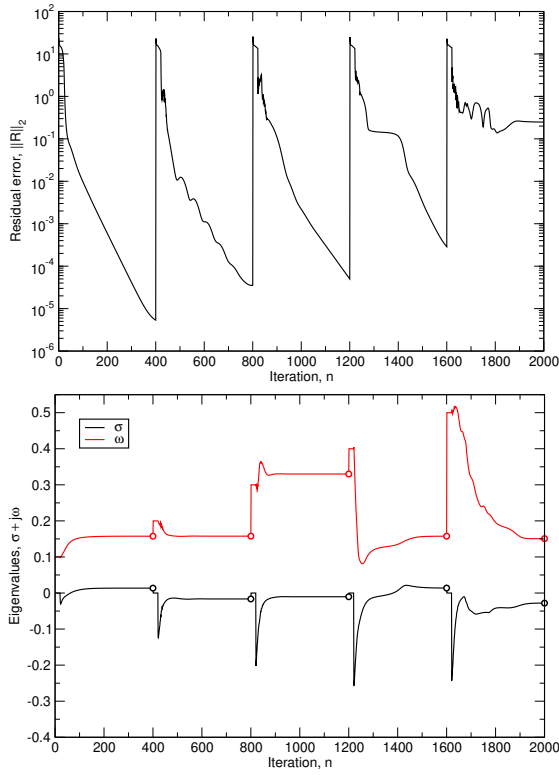


FIGURE 7: Top: convergence history of the real and imaginary parts of the five computed eigenpairs for the cylinder cascade case. Bottom: convergence of residual norms.

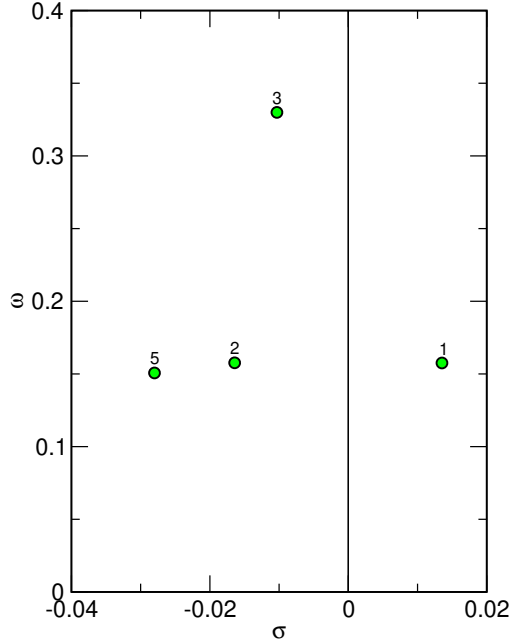


FIGURE 8: Computed eigenvalues $\lambda = \sigma + j\omega$ for the cylinder cascade case. The five points correspond to the first five eigenpairs computed using the method in Algorithm 2.

the inlet, and static pressure fixed at the exit—leading to standing pressure waves. In these modes, the pressure component is much larger relative to velocity than in mode one, which is velocity-dominated. The pressure contours are not plotted to the same relative scale as velocity in the three eigenmode plots.

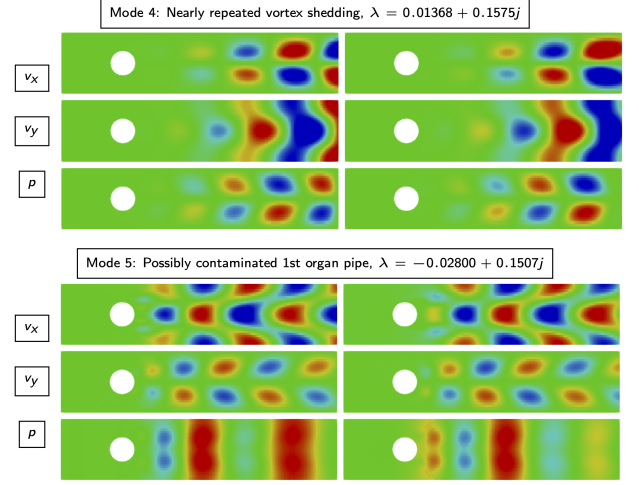


FIGURE 9: Pressure and velocity contours for the spurious fourth and fifth eigenmodes for the cylinder cascade case.

3.2 Purdue Rotor2

In this section, we compute several of the low-frequency eigenvalues and corresponding eigenmodes of unsteady flow in Rotor 2 of the Purdue three-stage compressor facility.

3.2.1 Two-dimensional Results. We first consider a two-dimensional annular slice of Purdue Rotor 2 [35] near the midspan of the compressor blades. This nearly two-dimensional analysis is useful for demonstrating the capability of the present method. For this model problem, air at standard conditions enters the rotor with zero swirl in the absolute frame. The back pressure was varied, and the steady flow was computed for a range of back pressures at a fixed rotor speed of 387.5 rad/s along the speedline. The computational grid used in this example has two cell layers in the radial direction, and 9,216 cells in each of the two radial layers. Figure 10 shows the computed performance, including the total pressure ratio across the row and the efficiency as a function of non-dimensional mass flow rate. Note that we were unable to compute converged solutions for back pressures greater than about 2195 lb/ft². Peak efficiency occurs at a back pressure of about 2180 lb/ft².

Next, we computed several of the lower frequency eigenvalues and corresponding eigenmodes for a back pressure of 2180 lb/ft². Shown in Fig. 11 is the eigenvalue convergence history for the zero nodal diameter case, along with the computed eigenvalues. Note that the convergence history is very similar to that found for the vortex shedding case, that is, the first two or three modes show good convergence, with later modes converging more slowly, or with stalled convergence.

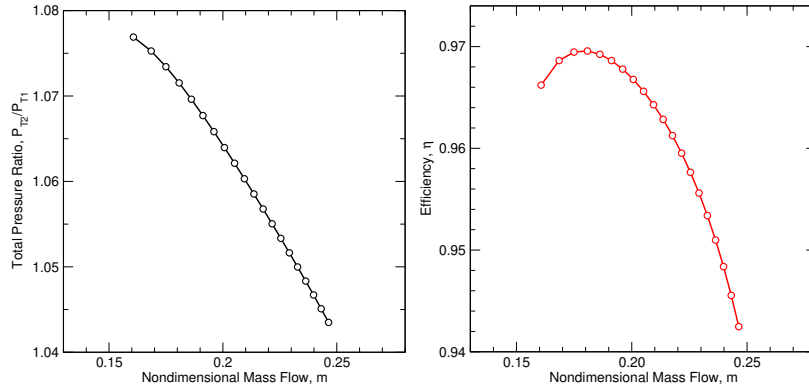


FIGURE 10: Computed performance for the two-dimensional section of Purdue Rotor 2 along the $\Omega = 387.5$ rad/s speed line. Total pressure ratio (left) and efficiency (right) versus nondimensional mass flow.

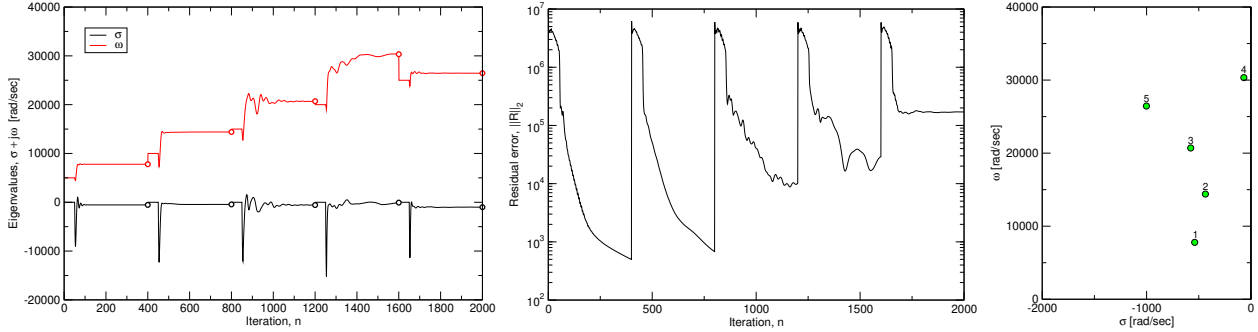


FIGURE 11: Left: convergence history of the real and imaginary parts of the five computed eigenpairs for the two-dimensional Rotor 2 case. Center: convergence of residual norms. Right: computed eigenvalues $\lambda = \sigma + j\omega$. Five eigenpairs were computed using Algorithm 2.

Shown in Fig. 12 are the first four computed mode shapes for the first four modes. These eigenmodes are analogous to so-called Parker modes. Parker computed the two-dimensional low-Mach-number pressure modes about an array of unstaggered flat plates in cross flow by solution of the acoustic wave equation, ignoring shed vorticity [17]. For the present analysis, on the other hand, the shed vorticity is clearly seen in the velocity component of the computed eigenmodes, with decreased wave length with increasing frequency as one would expect.

We next computed the two smallest eigenvalues and eigenmodes at the same steady operating condition for nodal diameters: 0, 3, 6, ..., 30 (equivalent to $-15, -12, \dots, 12, 15$). The converged eigenvalues for these eleven nodal diameters are shown in Fig. 13, annotated by nodal diameter number. All computed eigenvalues have negative real parts, indicating global linear stability—at least for this two-dimensional model. Note however that the frequency and damping of the eigenvalues changes significantly as the nodal diameter (interblade phase angle) varies.

3.2.2 Three-dimensional results. Next we compute a few of the low-frequency eigenpairs for the fully three-dimensional Purdue Rotor 2 geometry. For the case considered here, the prescribed static pressure at the compressor exit is 2140 lb/ft² and the rotor speed is 387.5 rad/s. For this (preliminary) computation, a total of the total 229,376 computational cells was

used for the single passage. (A relatively coarse grid was used here for expediency.)

Figure 14 shows the computed static pressure on the outer boundary of the computational grid. The structured computational grid outline is also shown. Note Rotor 2 has a tip clearance equal to 3 percent of the height of the constant radii annulus. The low-pressure region (in blue) emanating from the suction side of the tip clearance is due to the tip leakage vortex.

Next we computed the low-frequency eigenvalues for this case over a range of nodal diameters; the first two low-frequency eigenvalues computed are shown in Fig. 15. The solution for this case is not fully converged, which results in a somewhat irregular root locus as the nodal diameter is varied. We are currently investigating the cause of this behavior. The corresponding pressure component of the eigenmodes is shown in Fig. 16. Note that the lower frequency mode (Mode 2 in this case) has a nearly constant shape from hub to tip, except in the vicinity of the tip. The higher frequency mode (Mode 1), on the other hand, has a single node line, approximately at the half span.

4 Summary and Conclusions

In this paper, we have developed a fast and efficient computational method for performing eigen-analysis of three-dimensional rotating flow instabilities in turbomachinery, including phenomena such as rotating stall and non-synchronous vibra-

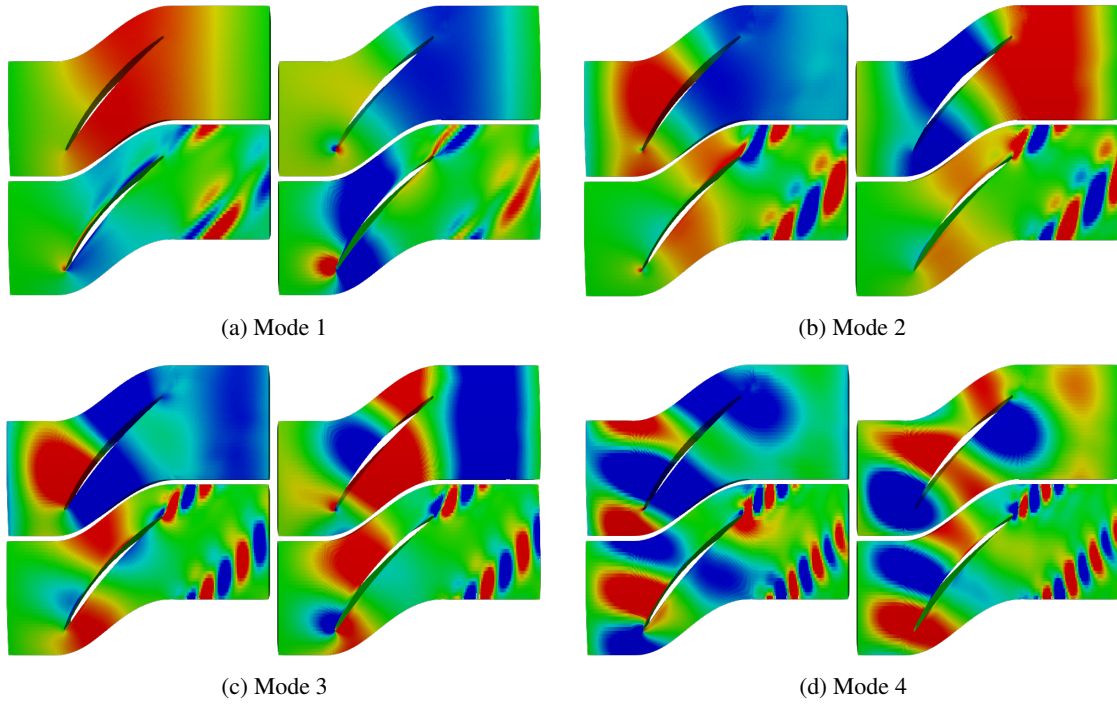


FIGURE 12: Eigenmodes of Rotor 2 flow. The left and right sides of each subfigure correspond to the real and imaginary parts of the eigenmode. The top and bottom corresponds to pressure and y-component of velocity.

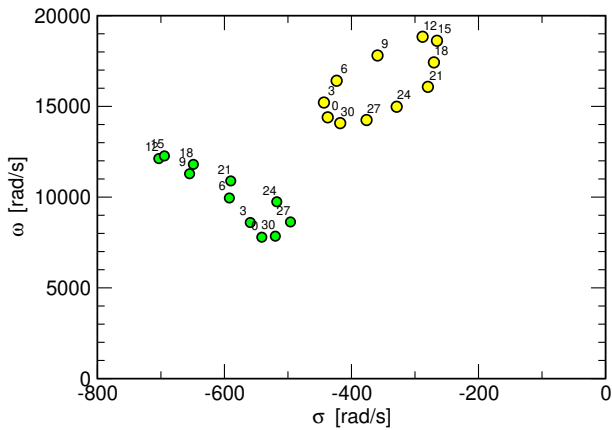


FIGURE 13: Root locus of the two smallest eigenvalues for the two-dimensional Rotor 2 case as the nodal diameter is varied.

tion.

To achieve this, we formulated a non-Hermitian, nonlinear eigenvalue problem derived from an existing harmonic balance (HB) CFD solver. The method is matrix-free, avoiding the need to explicitly form and store the Jacobian of the HB residuals. By applying harmonic linearization to both the HB residuals and the corresponding CFD fixed-point iterator, we are able to define the matrix–vector products required by the eigensolver. A small number of dominant eigenpairs can then be computed efficiently using an iterative eigensolver, with multiple eigenpairs obtained through soft deflation—without requiring the computation of left eigenvectors.

Because the method uses a linearized version of the original

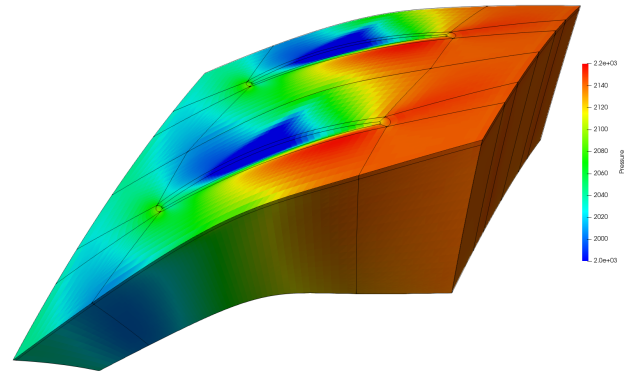


FIGURE 14: Computed steady pressure on the bounding surface of the computational grid for Purdue Rotor 2 (two passages shown for clarity). Note the wedge-shaped low-pressure region on the casing surface associated with tip leakage.

nonlinear HB flow solver as a preconditioner, it is computationally very efficient. Each eigenpair requires only about three to four times the computational cost of a corresponding steady flow calculation, making the approach practical for routine use on fully three-dimensional geometries. This is close to the theoretical limit of efficiency for matrix-free methods. In terms of iteration count, the convergence rate for the linearized HB equations is generally comparable to that of the steady solver, and may be slightly slower due to the time-spectral source term, which restricts the CFL number. There is also a small additional overhead associated with the finite-difference approximations used to evaluate matrix–vector products. Overall, the computational cost per eigenpair is typically about 3.5–4 times that of a steady solve,

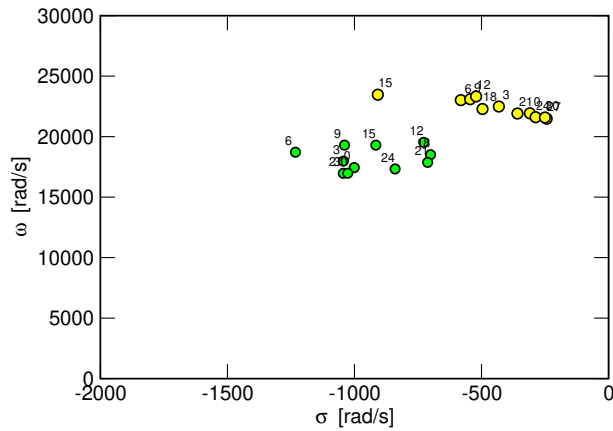


FIGURE 15: Root loci for two of the low-frequency eigenvalues of Purdue Rotor 2 as the nodal diameter is varied.

and the efficiency of the approach is governed by the same flow-acceleration techniques commonly used in *steady* flow solvers. A useful side benefit is that any improvements made to the steady flow solver will generally carry over to the eigensolver.

The method is designed to solve nonlinear eigenvalue problems directly, allowing non-reflecting boundary conditions to be applied—rather than the time-invariant inflow and outflow conditions commonly used in earlier work. Moreover, the method can be incorporated into existing harmonic balance solvers with minimal effort: in our case, fewer than 1,000 lines of Fortran code were required.

The proposed HB-based nonlinear eigenvalue problem (HB-NEP) solver provides a practical and efficient way to analyze the onset of fully three-dimensional flow instabilities in turbomachinery. It is straightforward to implement in existing harmonic balance codes and has a computational cost comparable to that of steady-state solvers, which makes it suitable for routine use. Future work will apply this method to study rotating stall and non-synchronous vibration in realistic three-dimensional turbomachinery configurations.

Acknowledgments

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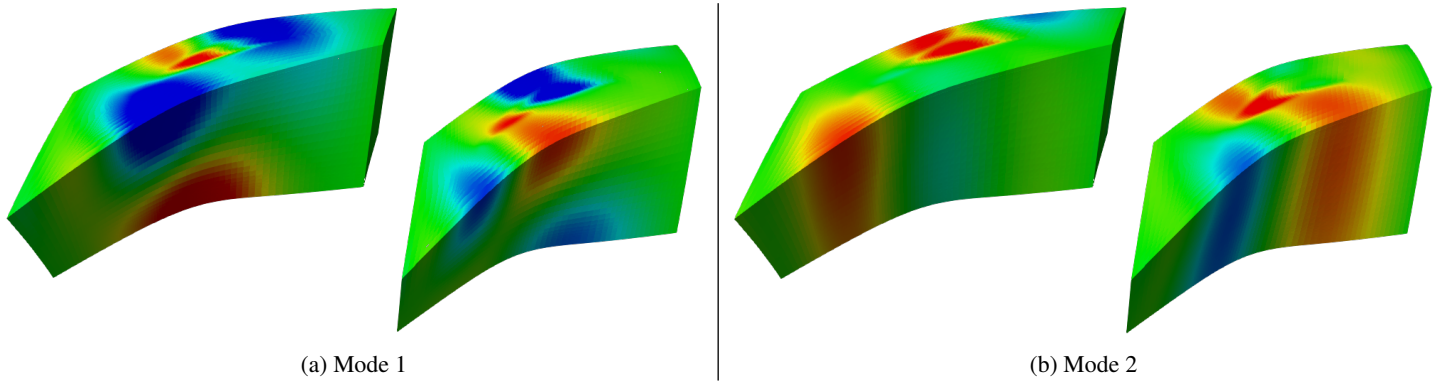


FIGURE 16: Pressure components of the two eigenmodes of Rotor 2 flow corresponding to the 6 nodal-diameter case shown in Fig. 15. Modes 1 and 2 are the higher- and lower-frequency modes, respectively. In each subfigure, the left and right images correspond to the real and imaginary parts of the eigenmode.

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